

Adroit Technologies

Protocol Driver

Name	Communications Link Serial Protocol Driver
DLL	D_COMLI_SER



COPYRIGHT

Copyright © 2022 Advanced Worx 112 (Pty) Ltd. All rights reserved.

Information in this document is subject to change without notice. The software described in this document is furnished under a license agreement or nondisclosure agreement. The software may be used or copied only in accordance with the terms of those agreements. No part of this publication may be reproduced, stored in a retrieval system, or transmitted in any form or any means electronic or mechanical, including photocopying and recording for any purpose other than the purchaser's personal use without the written permission of Advanced Worx 112 (Pty) Ltd.

Advanced Worx 112 (Pty) Ltd
20 Waterford Office Park
189 Witkoppen Road (c/o Waterford Drive)
Fourways
South Africa

<https://adroittech.co.za/>

Contents

1. Overview	4
1.1. Test Hardware	4
2. Wiring	5
2.1. DB9, DTE to DTE cable diagram RS-232C - 3 wire	5
2.2. DB9, DTE to DTE cable diagram RS-232C - 3 wire	5
2.3. DB9, DTE to DTE cable diagram RS-232C - 7 wire	5
2.4. RS422/485 – 2 wire, bus cabling	6
2.5. RS422/485 – 4 wire, bus cabling	6
3. Device Configuration	7
3.1. Device Configuration	7
3.1.1. PLC Details	8
3.1.2. Serial	9
3.1.3. Purging	9
3.1.4. Error handling – Multi-dropped system variables	9
3.1.5. Latency	10
3.1.6. Data Ranges	10
4. Supported Addresses	11
5. Protocol Definition and Considerations	12
5.1. Timesync Considerations	12
5.2. COMLI – Protocol function supported	12
6. General Notes and Troubleshooting	13
6.1. Contact Details	13
6.2. Adroit Technologies Knowledge Base	13
7. Appendix A - Configuring Mitsubishi FX5 as a COMLI slave	14
7.1. COMLI Function Block Setup	14
7.2. PLC program	14
7.3. PLC Wiring	15
7.4. Resulting Output	15
7.5. FX5 / COMLI I/O Mapping	16
8. Revisions	17
8.1. Document Revision	17
8.2. DLL Revision	17

1. Overview

This driver supports the ABB Communications link driver.

The 2nd Generation driver allows for the following features:

- Improved performance due to optimizations on mixed I/O to the same protocol area being transacted only once removing redundant transactions.
- Serial drivers by design don't benefit from multiple channels thus only one channel per device is enforced. Up to 247 Multi-dropped devices are supported per port.
- Demand scanning support, where read only I/O that is not on a mimic / graphic form and not logged or alarmed will be removed from the scanning system, the applicable tasks will dynamically reduce its size and merge into fewer scan tasks.

For improved network performance and reliability, it is recommended to use of industrial USB serial ports or true serial cards into the machines where applicable. RS-232 and RS-485 and RS-422 is supported.

Windows traditionally expects COM ports to always be available, therefore losing contact with Ethernet/serial extenders can cause lockups. Please contact Adroit support if you suspect this.

1.1. Test Hardware

This driver has been tested on the following hardware:

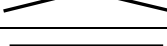
FX5U -32MT/DSS PLC using USER_FB_Comli_FX5U function block dated 2019/04/24 14:25:42

2. Wiring

Control of DSR/DTR CTS/RTS signalling is available as needed.
RS-232-c/RS-422/485, 4 and 2 wire supported.

2.1. DB9, DTE to DTE cable diagram RS-232C - 3 wire

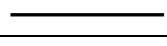
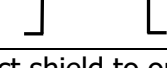
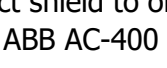
(With loopbacks in case the system expects signalling)

PLC 9 pin		PC 9 pin
2 RX		2 RX
3 TX		3 TX
5 GND		5 GND

Note: Connect shield to one side, PLC side only.

2.2. DB9, DTE to DTE cable diagram RS-232C - 3 wire

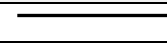
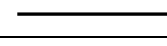
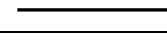
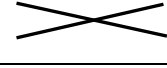
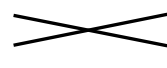
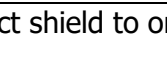
(With CTS/RTS loopbacks in case the system expects signalling, and DSR, DTR pulled high/Low on CD)

PLC 9 pin		PC 9 pin
2 RX		2 RX
3 TX		3 TX
5 GND		5 GND
7 RTS		7 RTS
8 CTS		8 XCTS
1 CD		1 CD
4 DTR		4 DTR
6 DSR		6 DSR

Note: Connect shield to one side, PLC side only.

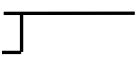
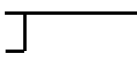
Also used for ABB AC-400 series DCS (diagram as above)

2.3. DB9, DTE to DTE cable diagram RS-232C - 7 wire

PLC 9 pin		PC 9 pin
2 RX		2 RX
3 TX		3 TX
5 GND		5 GND
7 RTS		7 RTS
8 CTS		8 CTS
4 DTR		4 DTR
6 DSR		6 DSR

Note: Connect shield to one side, PLC side only.

2.4. RS422/485 – 2 wire, bus cabling

PLC		PC
SG	————	Signal ground
SDA RDA		R+(D+)
SDB RDB		R-(D-)

*Polarity is important

2.5. RS422/485 – 4 wire, bus cabling

PLC		PC
SG*	————	Signal ground
SDA	————	R+
SDB	————	R-
RDA	————	T+
RDB	————	T-

*Ground can be left disconnected

*Polarity is important

3. Device Configuration

3.1. Device Configuration

Comli Serial Driver (2nd Generation) - CL5

☐ Enable secondary ☒ Primary ☐ Secondary

Primary connection parameters

☐ Enable shadow scanning from master to standby ☒ Enable standby reads
☐ Enable standby writes

Property	Value
Error handling	
Retries	3
Backoff count	5
Backoff period	20000
Disconnect after every transaction	False
Disconnect after every failure	False
Latency	
Inter station latency	0
Transaction latency	0
TXtoRX latency	0
Serial	
Com port	1
Baud rate	9600
Data bits	8 Databits
Parity	Even
Stop bits	1 Stopbit
Transaction timeout	3000
RS485 reflections	False
Purging	
Purge in standby	False
Purge type	Soft
Signaling	
DTR	Disable
DSR	Disable
RTS	Disable
CTS	Disable
PLC details	
Station ID	1
PLC Model	FX5
Debug level bitmask	960
Stale threshold	0
Max PDU	32
Data ranges for I/O bits (Decimal indexed)	
P	16376
Data ranges for I/O registers > 0x4000 (Decimal indexed)	
PR	3071
R registers (future)	
R	65535

Close

Enable secondary – Turn this option to have the driver maintain 2 channels of communication, the radio buttons **Primary** and **Secondary** can then be used to toggle between configurations.

Enable shadow scanning from master to standby – NASA specific function – per device agent level shadow scanning.

If enabled, writes will be sent to the master to be executed on its corresponding device and reads from master will be replicated to the standby.

If disabled, writes **Parallel scanning** is enabled.

Enable standby reads – If disabled, the standby will be considered cold, no reading will take place on the standby connection, and neither will any replication from the master occur.

Enable standby writes – If disabled, no writing will take place on the standby connection, and neither will any request be forwarded to the master.

3.1.1. PLC Details

Station ID – (Default = 1) PLC station id 1 – 247. Master is always 0.

PLC Model – (Default = ABB CPU) Choose a specific model.

Debug level bitmask – (Default: 480) - This is a bit mask that enable extra debugging in the datascope output, do not leave it enabled during normal operation on high-speed drivers as it will slow down the system depending on speed requirements. In serial drivers this delay is negligible. (Remember: All NASA drivers one can stop and start the device to reload these flags.)

960 default is just the protocol messages 40hex + 80hex+ 100hex + 200hex = 3C0hex -> 960 decimal in the dialogue.

#define DEBUG_MASK_COMSRESOURCE_INFO	0x00000001
#define DEBUG_MASK_COMSRESOURCE_CRITICAL	0x00000002
#define DEBUG_MASK_PROTOCOL_ITEM_VALUES	0x00000004
#define DEBUG_MASK_PROTOCOL_ITEM_BAD_VALUES	0x00000008
#define DEBUG_MASK_NASA_POOLS	0x00000010
#define DEBUG_MASK_TIMING	0x00000020
#define DEBUG_MASK_PROTOCOL_RESPONSES	0x00000040
#define DEBUG_MASK_PROTOCOL_REQUESTS	0x00000080
#define DEBUG_MASK_PROTOCOL_READ_DESC	0x00000100
#define DEBUG_MASK_PROTOCOL_WRITE_DESC	0x00000200

Stale threshold – (Default = 5) Every nth occurrence of an unchanged item value will be sent to the master. If the value is different, the value will be sent regardless of this count. This count reduces unnecessary event within the agent server. Set to report every poll value regardless of change, where 0 is not a valid option.

Max PDU length – (Default = 32) The overall maximum transaction length. 1 is the lowest value up to 32 × 16-bit registers or 32 × 16 discrete bits digitals per transaction.

3.1.2. Serial

Com Port – Only insert the number for the port. For example, “4” for COM4.

Baud rate – (Default = 9600) Select the required baud rate from the selection provided.

Data bits – (Default = 8) Select the required data bit count from the selection provided.

Parity – (Default = Odd) Select the required Parity mode from the selection provided.

Stop bits – (Default = 1) Select the required stop bit count from the selection provided.

Transaction timeout – (Default = 3000ms) Insert the required timeout for the system in milliseconds.

485 Reflections – (Default = False) Check if reflection detection and compensation is required on the port [**Currently not implemented**].

3.1.3. Purging

The driver will purge TX and RX buffers (data on the port) by default when required.

However, some systems such as Ethernet to serial port servers can become overloaded or even lockup if left receiving bytes in its internal buffer without a driver reading anything for a long period. For example, when a cluster partner is in standby, use the setting below to compensate:

Purge port in standby – (Default = False) Check if purging of the buffers is to be performed in the background while the cluster partner is in standby.

Purge mode – (Default = SOFT) **SOFT** cause a simple flushing via reading any available bytes, **HARD** sends a system call to tell the device itself to flush. Some systems have become unresponsive in HARD mode.

3.1.4. Error handling – Multi-dropped system variables

Retries – Enable this button to have the driver keep the PLC connection alive. Please see the PLC program and heartbeat sections below. This option is only useful with the watchdog program running in the PLC.

Backoff count – (Default = 5) Require 5 full consecutive failures including retries.

E.g. if 3 retries is configured and the **backoff count** is 5, the system will go into a backoff delay period in which this device will not be polled or written to for the Backoff period, in this example after the $((1 + 3) \times 5)$ consecutive failures.

Backoff period – Specify the millisecond interval the device will stay silent on the channel. This is very useful in multi-dropped systems as failing devices do not severely affect online devices.

Disconnect after every transaction – (Default = False) Tell the channel to close the port after every transaction. This can be useful on gateways if required.

Disconnect after every error – (Default = False) Tell the channel to close and reopen the port on every error during retries else it will only occur after the final retry.

3.1.5. Latency

Inter-PLC latency – The period that the system will wait before transmitting any read request or write request in multi-dropped system when a new device is contacted.

Transmission latency – The period that the system will wait before transmitting any read request or write request for the same PLC device. Inter message delay.

Tx to RX latency – The period that the system will wait after transmitting a request and before attempting to read the response.

3.1.6. Data Ranges

P – (Default = 16376) I/O bits area. Bits below 4000 hexadecimal index is in bits. Example: P 6, P 7, ... P n.

PR – I/O bits above 4000 hexadecimal (addressed in octal, index is in 16-bit registers. Example: PR 0 W, PR 1 W ... PR n W)

As of Rev 13 PR area now supports Bits in Registers as well. For example, PR 1000.0 the bit index is in decimal. 0 -> 15

R – Registers {assumed these are referring to the high registers}

A – Analogue outputs [**Currently not implemented**]

AI – Analogue inputs [**Currently not implemented**]

T 0 – Time Sync can be scanned to any agent slot but must be **Input disabled** and **Output enabled**. This is used to send down master time to the outstation.

4. Supported Addresses

ABB COMLI PLC data (device) types supported in the Adroit Protocol. Please note Decimal is used throughout.

Device	Prefix	Max Range	PLC Data Type	Boolean	Integer	Real	String
P – I/O bits	P	0 - 16376	Bit	P 0	-	-	-
(Octal)			Word16 Int16	- -	P 0 W P 0 I	P 10 W P 70 I	- -
PR – I/O registers	PR	0 – 3071	Bit *	PR 0.0 PR 0.15 PR 0 W	-	-	-
(Decimal)			Word16 Int16	- -	PR 0 W PR 79 I	PR 1 W PR 89 I	- -
T – Time sync	T	0	Bit	T 0	-	-	-
(write only, input inhibited)			Word16	-	T 0 W	T 0 W	-
R (Future)	R	0 - 65535	Int16	-	-	-	-
			Word16	-	-	-	-
			DWord32	-	-	-	-
			Long32	-	-	-	-
			Float32	-	-	-	-
			Int16	-	-	-	-
			String	-	-	-	-
A AI (Future)	A	-	Int16	-	-	-	-
			Word16	-	-	-	-
			DWord32	-	-	-	-
			Long32	-	-	-	-
			Float32	-	-	-	-
			String	-	-	-	-

* PR 0 W as a whole word can be scanned into a digital so that any value is deemed 1 (true) and 0 is 0 (false).

5. Protocol Definition and Considerations

5.1. Timesync Considerations

Scan address T 0, T 0 W scan to any discreet agent type such as BOOLEAN, INTEGER or REAL Agent.

Make sure the **Input disabled** and **Output enabled** check boxes are checked, as this type is write-only.

Tag scanning configuration : COMLI4

Device: COMLI4 (Started) OK

Agent: I2 Find Help

Slot: value (Header slots) Details...

Address: T 0 W Find Import...

Scan rate (ms): 4000 Deadband: 0

☒ Input disabled ☒ Output enabled

☐ Demand scan ☐ Output initialize

Currently scanned tags: Scan Unscan

Tag	Address	Rate	Deadband	IO flags
I1.value	P 40 W	4000	0	3
I2.value	T 0 W	4000	0	2
_SAMPLEDIG...	P 0	4000	0	3

The driver will then send down the system time in UTC time format.

5.2. COMLI – Protocol function supported

- 0 30H (Write) Transfer Multiple I/O bits (P 0) or register (P 0 W)
- 1 31H Acknowledge.
- 2 32H Requests multiple I/O bits or registers groups of at least 8 bits, registers are groups of 16bits
- 3 33H (Write) Transfer single I/O bit
- 4 34H Request a single I/O bit. (Redundant function 2 is used)
- J 4AH Transfer date and time in UTC format.

If more messages are required, please contact the Adroit driver department as these will be added as required.

6. General Notes and Troubleshooting

6.1. Contact Details

Email: support@adroit.co.za and drivers@adroit.co.za

Phone: +27 11 658-8100

Web: www.adroit.co.za

6.2. Adroit Technologies Knowledge Base

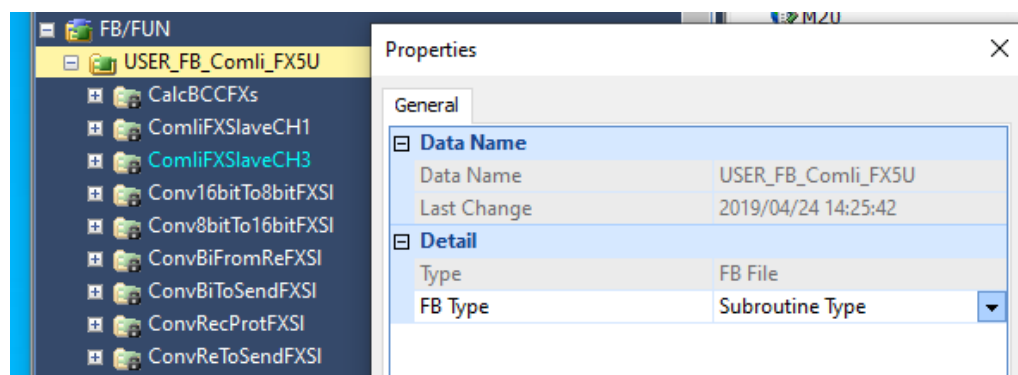
For additional information, please refer to our Knowledgebase website:

[Latest Drivers topics - Features, discussions, tips, tricks, questions, problems and feedback \(adroittechnologiesautomation.com\)](http://adroittechnologiesautomation.com)

7. Appendix A - Configuring Mitsubishi FX5 as a COMLI slave

7.1. COMLI Function Block Setup

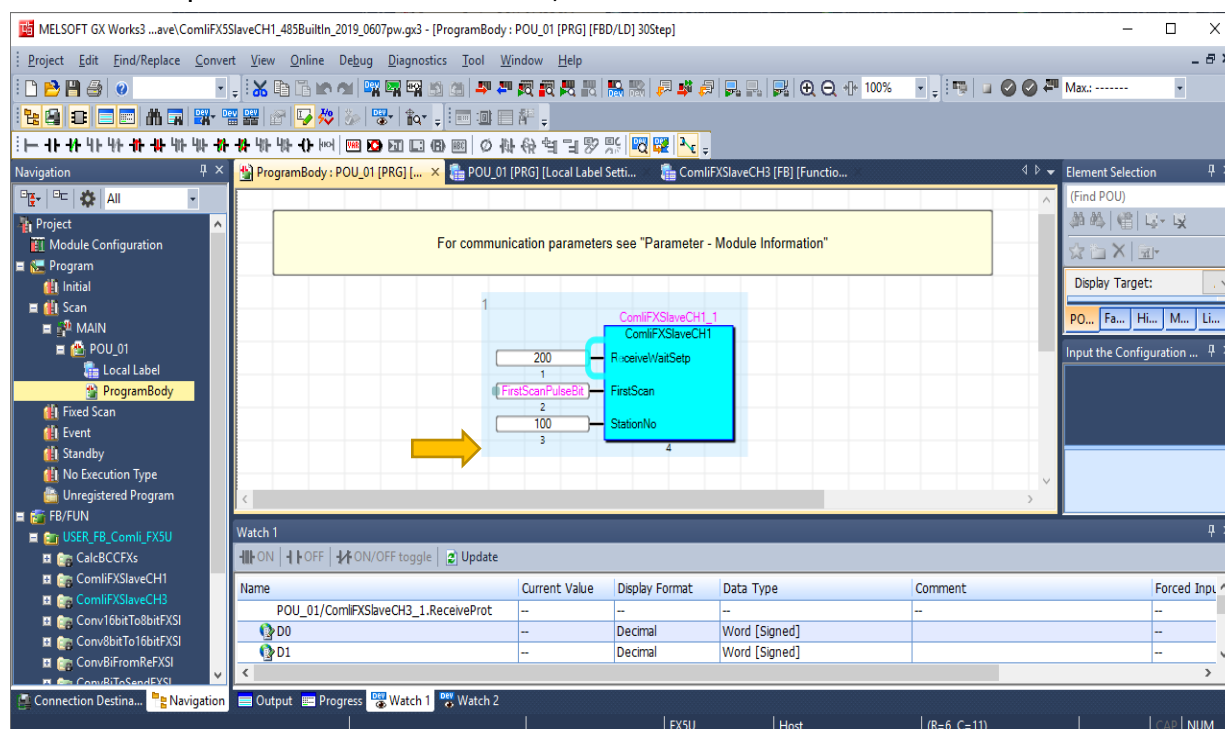
This example uses the FX5U -32MT/DSS Plc using CH1 of USER_FB_Comli_FX5U function block dated 2019/04/24 14:25:42



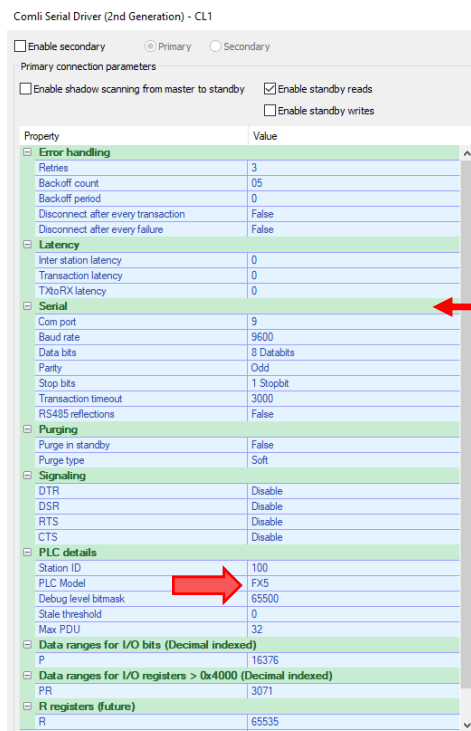
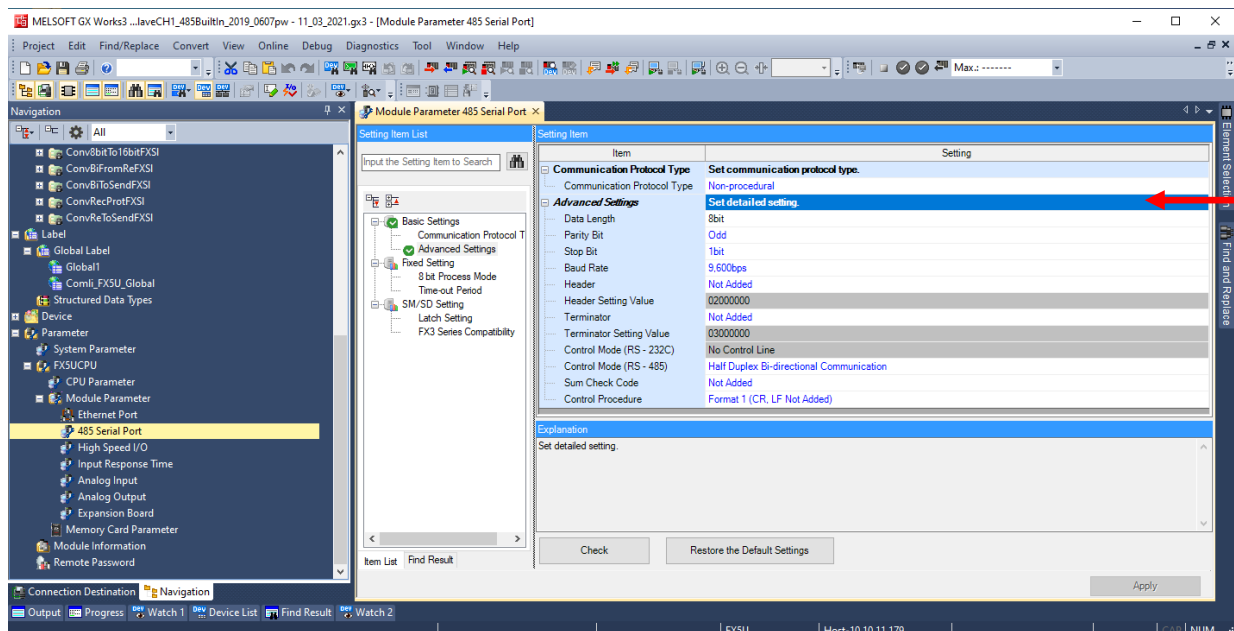
7.2. PLC program

Fx5 plc with a serial port, in this example a FX5U-32MT/DSS was used.

The RS-485 port on the FX5 is channel 1, therefore ComliFXSlaveCH1 function block must be used.



Make sure the COM port settings match the Master driver settings. Baud rate, Databit length, stopbits and parity. Check that the station number matches what is required.



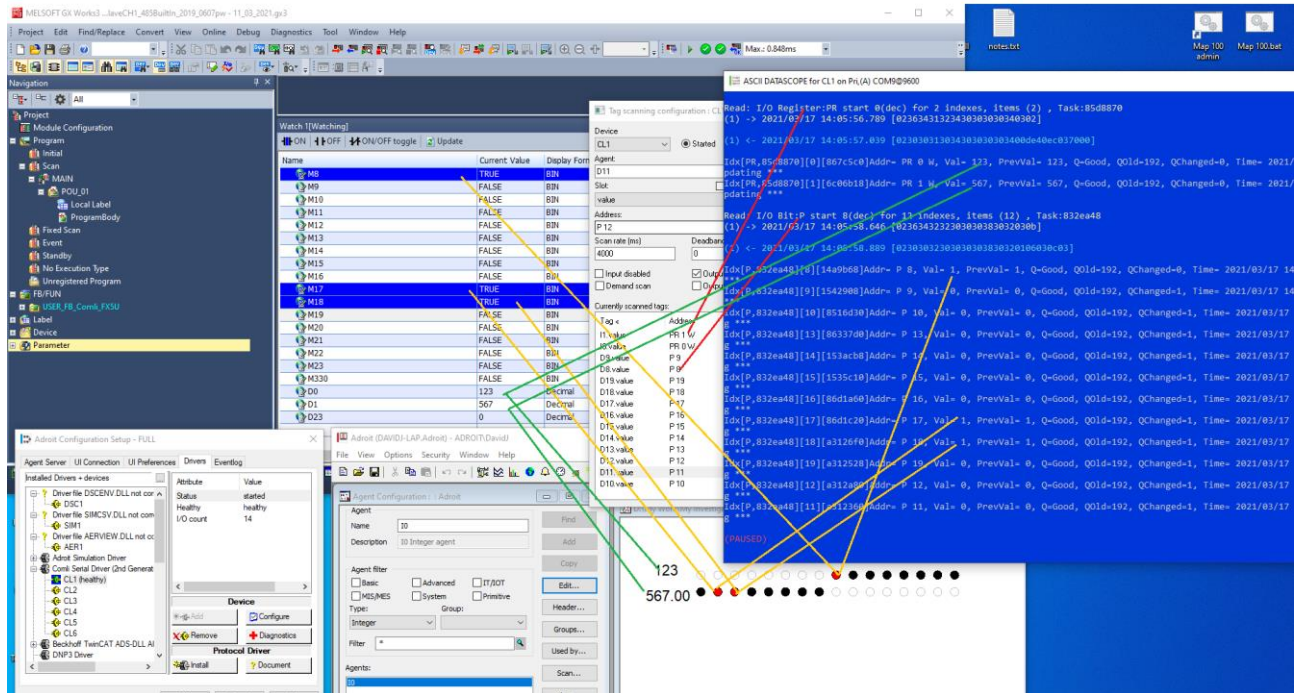
7.3. PLC Wiring

Example provided is wired for RS485 (2-wire)

Hint: use 2 terminal programs and a second RS485 port, make sure to test the link by sending and receiving on the opposite end in both directions. This confirms the TX and RX buffers are working on the 485 modules.

7.4. Resulting Output

Resulting montage PLC monitored via gxworks3, Adroit classic UI, Adroit datascope below.



7.5. FX5 / COMLI I/O Mapping

- COMLI P's are described in OCTAL in the COMLI protocol specification, but we have decided to prevent confusion to rather standardize as decimal.
- P's map to M's in the FX5 plc ... P0 ~ M0
- P 0 W in Adroit is a 16-bit array or P's mapped to COMLI P0, P1 ... P15 ~ [M0, M1 ... M15]
- PR's in COMLI are any 16-bit register addressing decimal higher than 0x4000 but addressed as decimal starting from 0 such that PR 0 W maps to D0 in Mitsubishi.
- PR 0 W ~ D0, PR 1 W ~ D1 ... etc
- PR 0.0 bit in Register read and writes (with read before write) to digital agents is supported.
- PR 0 W as a whole word can be scanned into a digital also so that any value is 1, 0 is 0.

8. Revisions

8.1. Document Revision

Rev.	Date	Author	Comment	Approval
1.0	14-Dec-2022	J. Duncan	New Format	D. Jordaan
1.2	19-Jan-2022	K. Robinson	Removed shadow effect on images, content formatting and changes. Changed arrow graphics.	D. Jordaan
2.0				
3.0				
4.0				
5.0				
6.0				

8.2. DLL Revision

Rev.	Date	Changes
14	17/03/2021	Updated documentation.
18	16/11/2022	Shadow mode partner (with offline redundant plc) in standby did not mark affected tags bad when becoming master.

